Bruno Henrique Nunes

Applied AI / Machine Learning Engineer

Brazil | Open to remote opportunities hiring from Brazil

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Professional Summary

Applied AI and Machine Learning Engineer with a completed MSc in Artificial Intelligence from UNICAMP. Develop AI agents, multi-agent workflows and LLM applications using RAG, NLP and prompt engineering. Combine Python/API engineering, LLM evaluation, automated testing and observability with reproducible deep-learning research and statistical analysis.

Technical Skills

Applied AI and LLM Engineering: Python, AI agents, multi-agent systems, large language models (LLMs), RAG, Natural Language Processing (NLP), prompt engineering, LLM evaluation, semantic search, text embeddings.

Machine learning and Data Science: PyTorch, deep learning, computer vision, statistical analysis, experiment design, confidence intervals, data quality and topic modeling.

Information retrieval: Evidence retrieval, vector search, semantic similarity and retrieval-augmented generation.

Software engineering: FastAPI, REST APIs, automated testing, observability, AWS Lambda, AWS SES, Git, Azure DevOps CI/CD.

Professional Experience

Application Developer — IBM Aug 2026 – Present

- Build Python-based AI agents and multi-agent workflows for applied AI, software engineering and Data Science use cases.
- Develop retrieval-grounded LLM applications with RAG, NLP, evidence retrieval, prompt engineering and Python/FastAPI REST APIs.
- Implement prompt-injection defenses, output guardrails, automated testing and observability; support application deployment.

Selected Research and Projects

Deep Learning for Prostate Histopathology Feb 2024 – Sep 2026

Master's research, University of Campinas (UNICAMP)

- Developed a research pipeline for binary segmentation of prostate cancer in histopathological images, using CNN/Transformer-based modeling and PyTorch.
- Implemented data preparation and quality control, patient-level data partitioning, model training, ensemble optimization and final evaluation.
- Preserved configurations, data provenance and experimental outputs to support reproducibility. The REDU deposit contains generated research artifacts; original DIAGSET images are not redistributed.

Code: github.com/bhnunes/Master_Coding | Artifacts: doi.org/10.25824/redu/EDPCF2

GEODITE — Equivariant Interatomic Potentials

Contributed to model implementation, training and evaluation for an equivariant message-passing model for interatomic potentials.

Co-author, *Equivariant Interatomic Potentials without Tensor Products* (arXiv preprint, January 2026).

arxiv.org/abs/2601.15492

Selected 2026 Applied AI and Data Science Projects

- **LLM evaluation (Feb–Mar):** Combined lexical and semantic signals, text embeddings and model-assisted judgment to evaluate response quality.
- **AI agent engineering and evaluation (May–Jul):** Built Python tooling for multi-agent software-engineering workflows with repository-aware guidance and automated tests; designed paired statistical analyses using the Wilcoxon signed-rank test, bootstrap confidence intervals and token usage metrics.
- **Applied Data Science and NLP (Jul–Oct):** Prototyped RAG-supported analysis, multilingual text classification and semantic search; applied data validation, topic modeling, statistical metric design, Apache Spark and telemetry enrichment with REST APIs, PostgreSQL and Kubernetes.

Teaching Experience

Teaching Assistant — UNICAMP Jul 2024 – Dec 2025

Supported more than 200 undergraduate students in database management and object-oriented programming; led problem-solving sessions in SQL, relational algebra and OOP and supported assessment. Internship appointment.

Selected Earlier Experience

Software Developer — PAteam (freelance) Jan 2024 – May 2024

Built Python data-ingestion automation using AWS Lambda, REST APIs and AWS SES.

Software Developer — Expleo Group (contract) May 2022 – Aug 2023

- Built Python/Robocorp automation and API integrations for a legacy-to-Salesforce migration using Salesforce REST APIs and SQL Server; documented workflows and supported Azure DevOps CI/CD, monitoring and issue resolution.

Senior RPA Developer — Nestlé Feb 2022 – Aug 2023

- Delivered 13 enterprise automations using Python, Blue Prism, PowerShell, SQL and Automation Anywhere; reviewed code across 10+ projects, mentored developers and supported change approval for more than 50 robots.

Senior RPA Developer — Deloitte Nov 2021 – Feb 2022

Translated stakeholder requirements into Python, Node.js and UiPath automation for a healthcare IoT integration project; presented solutions to clients and suppliers.

IBM — Earlier roles

RPA Developer: Oct 2019 – Nov 2021 | Software Developer: Oct 2018 – Oct 2019

IT Analyst: Jan 2017 – Oct 2018 | Support Specialist: Jan 2012 – Oct 2014

- Delivered automations and integrations with Python/Selenium, APIs, SAP, Db2, Cloudant, Blue Prism, VBA and JavaScript.
- Created invoice analytics with VBA and Power Query for 20,000+ records and 10+ validation KPIs; reported processing time fell from more than three hours to about six minutes.
- As Support Specialist, provided English-language support to US and Canadian clients and used SQL for compliance checks and billing-adjustment reporting.

Education

MSc in Artificial Intelligence — University of Campinas (UNICAMP) 2024 – 2026

Completed September 2026; research in deep learning for prostate histopathology.

MBA in Computer Software Engineering — FIA Business School 2022 – 2023

Bachelor's in Engineering Physics / Applied Physics — University of São Paulo 2006 – 2010

Selected Certifications

AWS Certified Generative AI Developer – Professional Oct 2026

AWS Certified Machine Learning Engineer – Associate Jul 2026

AWS Certified Developer – Associate Oct 2025

AWS Certified Solutions Architect – Associate Apr 2025

Languages

Portuguese: Native. **English:** Full professional proficiency; TOEIC 890 (Sep 2025).